Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Variance Decomposition Across Five Engineering Choices

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July 28, 2026

Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by four orders of magnitude. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. I hold a single spur gear pair (module 3 mm,20 teeth,42CrMo4/AISI 4140,700 N·m,constant amplitude, $R = 0$) fixed and systematically vary five methodologicalchoices: size-and-surface factor formulations drawn from three standards (ISO 6336, AGMA 2001-D04, FKM 7th ed.), surface roughness grade (Rz = 4, 15, 50 μm), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). The tooth root stress concentration is handled by the ISO 6336 Y_{Sa} factor within the nominal stress calculation; a separate fatigue notch factor K_f was removed to avoid double-counting of the notch effect. A full-factorial AFT survival model over all 144 combinations (including 28 run-outs handled via censored likelihood) decomposes the total log-variance. At the reference torque of 700 N·m—near the endurance limit for this gear, where the $\sigma_m/\sigma_{\text{uts}}$ ratio is ~0.39—three factors share the dominant contribution: mean-stress correction (33.9%), size-and-surface standard (30.1%), and surface roughness (24.6%). A Standard × Rz interaction accounts for 3.0%. The S–N curve source (7.5%) and cumulative damage rule (0.9%) are secondary. Bootstrap validation confirms stable rankings.These percentages are specific to the high-torque condition studied here. The factor ranking is operating-point-dependent: at lower torque the standard choice dominates, while mean-stress correction and surface roughness rise to dominance near the endurance-limit regime explored here. The analysis is confined to quenched-and-tempered steel without residual stresses; in case-carburized gears, the compressive residual stress field would substantially reduce the mean-stress contribution below the 33.9% reported here. Whereas prior studies—including recent work in this journal [12]—ask whether two standards produce different predictions, this study asks how much each methodological switch contributes to total log-variance relative to all others, by simultaneous factorial decomposition across five dimensions. No prior work has ranked the sources of method-induced divergence in gear fatigue life prediction by variance explained. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex).All formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.); S–N parameters are from the publicly accessible Waterloo SMDIdbase.

Keywords: gear fatigue, uncertainty quantification, AFT survival analysis, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear's fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox is designed by an engineer consulting a standard, selecting a correction method, and computing a life estimate. Different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

The gear fatigue literature has long acknowledged that methodological choices matter [4, 5]. Studies have compared S–N curves from different databases [8], evaluated mean-stress correction methods [6], and benchmarked cumulative damage rules [7]. Each study examined one factor in isolation while holding others at default values. No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

The variance-decomposition methodology is a standard tool for identifying dominant sources of divergence when different analysis pipelines are applied to identical input data. Here, I apply this framework to gear fatigue life prediction.

I address two questions: which methodological choices produce the largest divergence in predicted gear bending fatigue life, and can a bootstrap noise floor distinguish genuine signal from finite-sample fluctuation?

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB. The loading is constant-amplitude pulsating bending ($R = 0$).

2.2 Five Methodological Switches

2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation $\sigma_a = \sigma'_f \cdot (2N_f)^b$ relates stress amplitude to cycles to failure. I use two experimentally-fitted parameter sets for quenched-and-tempered AISI 4140 ($\sim$300 HB), drawn from the publicly accessible University of Waterloo SMDIdbase [8]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, $\sigma_e = 500$ MPa (staircase-method endurance limit)

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ultimate tensile strength	1000	MPa
Input torque	700	N·m
Speed	1500	rpm
Stress ratio, R	0	—

- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, $\sigma_e = 550$ MPa

The 328 MPa spread in σ'_f and the 50 MPa difference in endurance limits conflate two distinct sources of variation: (i) intrinsic material scatter between different heats, specimen batches, and testing laboratories, and (ii) differences in S–N curve fitting methodology (#68 used the staircase method for endurance limit determination; #66 used a different fitting protocol). The 7.5% S–N source variance reported in Table 2 therefore captures the combined effect of material variability *and* fitting-method choices, not material scatter alone. A factorial separation of these two sub-sources would require additional S–N datasets with documented fitting histories. The raw data are freely downloadable; no proprietary or subscription-gated sources are used.

2.2.2 Switch 2: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

- **Goodman:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma_{\text{uts}})$
- **Gerber:** $\sigma_{ar} = \sigma_a/(1 - (\sigma_m/\sigma_{\text{uts}})^2)$
- **Morrow:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma'_f)$, where σ'_f is taken from the active S–N curve
- **SWT:** $\sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a}$

2.2.3 Switch 3: Cumulative Damage Rule (2 options)

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.4 Switch 4: Size and Surface Standard (3 options)

- **ISO 6336-3:** $Y_X = 1.0$ (size factor, $m_n \leq 10$ mm). $Y_{R\,\text{rel}\,\text{T}} = 1.674 - 0.529(R_z + 1)^{0.1}$ (relative surface factor, for quenched & tempered steels, valid for $0.5 \leq R_z \leq 40$ μm; capped at $R_z = 40$ μm value of 0.907 for rougher surfaces). $Y_{R\,\text{rel}\,\text{T}} = 1.120$ for $R_z < 1$ μm.

- **AGMA 2001-D04:** $K_s = 1.0$ (size factor, Clause 15, Figure 17; diametral pitch $= 8.47 \geq 5$). $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$ (Clause 16, Eq. 16.1), where R_f is R_a in μin and S_{ut} in ksi. $R_z \to R_a$ via $R_a \approx R_z/6$.

- **FKM Guideline (7th ed., Section 4.4.2):** $K_d = 1 - 0.05 \cdot \log_{10}(m_n \cdot 10)$ (statistical size factor for wrought steel gears with $m_n \leq 10$ mm; the FKM guideline provides a more general table-based procedure for other geometries). $K_{F\sigma} = \max(1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min}), 0.55)$, with $R_{m,N,\min} = 400$ MPa for wrought steels.

2.2.5 Switch 5: Surface Roughness (3 options)

- **Ground, Rz = 4 μm:** Precision aerospace quality
- **Machined, Rz = 15 μm:** Standard industrial quality
- **As-forged, Rz = 50 μm:** Rough, agricultural machinery

2.3 Stress Concentration Treatment

The ISO 6336-3 Method B nominal stress (Eq. 1) includes the stress correction factor $Y_{Sa} \approx 1.55$, which converts the nominal bending stress to the local peak stress at the tooth root fillet. Because Y_{Sa} already accounts for the geometric stress concentration, applying a separate fatigue notch factor K_f (e.g. via Peterson or Neuber) on top of σ_{F0} would double-count the notch effect. The ISO 6336 framework handles notch *sensitivity* through the relative notch sensitivity factor $Y_{\delta\,\mathrm{rel\,T}}$, which reduces the *allowable* stress rather than amplifying the *applied* stress. Since $Y_{\delta\,\mathrm{rel\,T}}$ is not among my switches, I omit K_f from the pipeline and rely on Y_{Sa} for the stress concentration. A third-party method for K_f (e.g. a 3D finite-element extraction of the notch-root stress gradient) could be added in future work without double-counting, provided the base stress is computed without Y_{Sa}.

This treatment is internally consistent for ISO 6336, where Y_{Sa} already converts the nominal bending stress to the local peak stress. For AGMA 2001 and FKM, however, the situation is asymmetric: neither standard employs a Y_{Sa}-equivalent coefficient in its native formulation, and both rely on independent notch sensitivity or stress-gradient-based approaches that are not replicated here. The uniform removal of K_f and adoption of Y_{Sa} therefore alters the native stress calculation chains of AGMA and FKM in two ways—introducing a factor they do not use and removing factors they do. The quantitative impact of this asymmetry on the standard-choice variance is assessed via sensitivity analysis in §4.5; it accounts for at most 2–6 pp of the reported 30.1% standard-choice variance, and the reported value should be interpreted as a lower bound on the true between-standard divergence.

2.4 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} Y_{Sa} Y_\varepsilon Y_\beta \tag{1}$$

where $F_t = 2T/d_1$ is the tangential force ($T = 700$ N·m, $d_1 = m_n z_1 = 60$ mm, giving $F_t \approx 23.3$ kN); $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (ISO 6336-3 Annex A, Figure A.1, for $z = 20$, $\alpha = 20°$, standard ISO 53 basic rack, no profile shift); $Y_\varepsilon = 0.25 + 0.75/\varepsilon_\alpha \approx 0.691$ with $\varepsilon_\alpha \approx 1.7$; and $Y_\beta = 1.0$ for spur gears. The reference condition gives $\sigma_{F0} \approx 778$ MPa.

2.5 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B includes six additional multiplicative factors: application factor K_A, dynamic factor K_V, face load factor $K_{F\beta}$, transverse load factor $K_{F\alpha}$, rim thickness factor Y_B, and deep tooth factor Y_{DT}. All six are set to unity—equivalent to assuming a perfectly smooth drive train, quasi-static loading, perfect tooth contact, and standard proportions. These assumptions are necessary to isolate the methodological switches from application-specific load uncertainties, but they constrain the absolute N_f values to an idealized gear pair. Whether industrial load factors ($K_A > 1$, $K_V > 1$, $K_{F\beta} > 1$) would alter the *relative ranking* of the methodological switches has not been tested here. Since higher load factors increase the effective stress, they would shift the operating point further above the endurance limit, which—given the torque-dependence documented in §3.5—could modify the variance shares attributed to mean-stress correction and surface roughness. A sensitivity analysis varying K_A and K_V across industrially representative ranges is left for future work.

2.6 Pipeline

For each of the $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations:

1. Compute σ_{F0} (Eq. 1)
2. Apply mean stress correction (Switch 2) $\rightarrow \sigma_{ar}$
3. Apply size and surface factors (Switch 4, Switch 5) $\rightarrow \sigma_{\text{eff}}$
4. Solve Basquin equation (Switch 1) $\rightarrow N_f$
5. Apply damage rule threshold (Switch 3) $\rightarrow N_f^{(\text{final})}$

2.7 Statistical Analysis

2.7.1 Log-Normal AFT with Censored Likelihood

I fit a log-normal accelerated failure time (AFT) model to $\log_{10}(N_f)$ with all five factors as categorical predictors. The 28 run-out entries are treated as right-censored observations contributing $\log P(T \geq t_{\max})$ to the likelihood, rather than being discarded. The variance fraction attributed to factor j is the proportional reduction in log-likelihood when that factor is removed from the full model: $\Delta \log \mathcal{L}_j \,/\, \sum_k \Delta \log \mathcal{L}_k \times 100\%$. This approach preserves all 144 observations and naturally decomposes variance without requiring the normality and homoscedasticity assumptions of ordinary least-squares ANOVA.

Residual diagnostics for the fitted AFT model are as follows. The residual distribution shows mild negative skew (skewness $= -0.44$) but the Shapiro–Wilk test rejects strict normality ($W = 0.960$, $p = 0.0016$). With 116 observed failure times, even modest departures from normality are statistically detectable; the practical question is whether the non-normality is severe enough to distort the variance decomposition. Levene tests for homogeneity of residual variance across factor levels confirm that the variance is stable for the three dominant factors (mean-stress: $p = 0.52$; standard: $p = 0.59$; roughness: $p = 0.62$), while the S–N curve source shows mild heteroscedasticity ($p = 0.004$). I proceed with the log-normal AFT as a working model, noting that the likelihood-ratio-based variance decomposition is more robust to moderate non-normality than F-tests would be, but the reported variance fractions should be interpreted with this caveat in mind (see also §4.5).

2.7.2 Bootstrap Noise Floor

I bootstrap-resample the 144 entries (with replacement) 2000 times (RNG seed 42), refitting the full AFT model and all drop-one-factor models on each draw. The signal-to-noise ratio $\mathrm{S/N} = \bar{\theta}_j / \sigma_{\mathrm{boot,j}}$ is the bootstrap analogue of a z-statistic; under approximate normality of the bootstrap distribution, $\mathrm{S/N} > 1.96$ corresponds to a 95% confidence interval that excludes zero. I adopt $\mathrm{S/N} = 3$ as a conservative reporting threshold (corresponding to $\approx$ 99.7% confidence), noting that this is more stringent than the conventional 1.96 benchmark. The bootstrap 95% confidence intervals (percentile method) are reported alongside the point estimates in §3.

3 Results

Table 2: Log-normal AFT variance decomposition—point estimates from the full model fit ($2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations; 28 run-outs handled via censored likelihood). Bootstrap means (2000 draws, RNG seed 42) differ slightly from these point estimates and are shown in Fig. 2.All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed. S–N parameters from Waterloo SMDIdbase.

Factor	ΔlogLik	Var.%	S/N
Mean stress correction	183.07	33.9	¿10
Size/surface standard	162.75	30.1	¿10
Surface roughness, R_z	133.08	24.6	¿10
S–N curve source	40.60	7.5	6
Standard $\times$ R_z	16.28	3.0	3
Cumulative damage rule	4.72	0.9	2
σ_{AFT}		0.184	—

3.1 Variance Decomposition

Table 2 presents the log-normal AFT variance decomposition, based on all 144 combinations with the 28 run-outs handled through censored likelihood (EM estimation). Three factors share the dominant contribution: mean-stress correction (33.9%, $\mathrm{S/N} \approx 25$), size-and-surface standard (30.1%, $\mathrm{S/N} \approx 25$), and surface roughness (24.6%, $\mathrm{S/N} \approx 22$). Together they account for 88.6% of total log-variance. Bootstrap confidence intervals from 2000 draws are narrow ($\pm$1–2 percentage points), confirming that all three are robustly above the noise floor.

The S–N curve source (7.5%, S/N $\approx$ 6) and the Standard $\times$ R_z interaction (3.0%, S/N $\approx$ 3) are secondary but distinguishable from zero. The cumulative damage rule (0.9%, S/N $\approx$ 2) is negligible in the constant-amplitude regime. The AFT scale parameter $\sigma = 0.184$ indicates that the model captures the dominant structure of the data; the remaining unexplained dispersion is modest.

This three-factor picture differs from the ordinary-ANOVA result reported in an earlier version of this work (which gave 31.1% standard, 30.9% mean-stress, 8.5% Rz). The difference arises because the 28 run-out combinations are predominantly associated with low surface roughness and high-endurance S–N curves; excluding them from the analysis—as ordinary ANOVA must—systematically underestimates the contribution of surface-finish-related factors. The censored-likelihood AFT framework eliminates this bias by retaining the partial information provided by run-out observations.

3.2 Mean Stress Correction

At the reference nominal stress ($\sigma_{F0} \approx 778$ MPa), the four correction methods diverge substantially: Gerber predicts the lowest median equivalent stress ($\sigma_{ar} \approx 458$ MPa), Goodman the highest ($\approx$ 636 MPa), with Morrow ($\approx$ 545 MPa) and SWT ($\approx$ 550 MPa) intermediate. This 39% spread in equivalent stress translates to a factor of $\sim$10–100 in predicted cycles for combinations above the endurance limit. The result contradicts the common assumption that mean-stress effects are negligible for gear bending at $R = 0$, and follows directly from the stress magnitude: at $\sigma_{F0} \approx 778$ MPa, the $\sigma_m/\sigma_{\text{uts}}$ ratio is $\sim$0.39, large enough for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible. The four methods are not equally appropriate for all materials: Goodman was developed for brittle materials and is conservative for ductile steels such as 42CrMo4; Gerber, Morrow, and SWT are better matched to ductile behavior. To assess how much of the reported mean-stress variance arises from this domain mismatch, I re-ran the AFT decomposition excluding Goodman. With the three ductile-appropriate methods (Gerber, Morrow, SWT), the mean-stress contribution drops from 33.9% to 26.1%, and the factor ranking becomes standard (32.7%), roughness (27.7%), mean-stress (26.1%). The three factors remain co-equal; approximately 8 pp of the mean-stress variance is specific to the Goodman-versus-ductile contrast, while the remaining 26 pp reflects genuine divergence among methods all considered appropriate for this material class. I retain all four methods in the main analysis because industrial practice—particularly in conservative design codes—does not restrict mean-stress correction to material-matched methods, and Goodman remains in wide use for gear steels despite its brittle-material origin.

This finding does *not* contradict the traditional engineering view that mean-stress effects are minor at $R = 0$ under typical operating conditions. Rather, the 33.9% figure is specific to the high-stress condition studied here. At 600 N·m ($\sigma_{F0} \approx 667$ MPa, $\sigma_m/\sigma_{\text{uts}} \approx 0.33$), the mean-stress contribution drops to approximately 10% under the earlier ANOVA framework (§3.5)—a threefold reduction over a 17% torque decrease. For the vast majority of industrial gearboxes, which operate at stress levels well below the endurance limit of the material, the mean-stress correction would contribute far less than the 33.9% reported here, and the standard choice or surface roughness would dominate the uncertainty budget. The contribution reported in Table 2 is therefore best understood as the *upper envelope* of mean-stress influence— realized only when the gear is intentionally pushed toward its endurance limit, as in weight-critical aerospace or motorsport applications. A quantitative mapping of mean-stress variance share as a function of $\sigma_m/\sigma_{\text{uts}}$

across a continuous torque range is left for future work.

I emphasize that this finding—and the 33.9% mean-stress variance fraction reported in Table 2—is specific to quenched-and-tempered steel without residual stresses. In case-carburized gears, which dominate automotive and aerospace applications, the compressive residual stress in the case layer (typically 200–400 MPa at the tooth flank) substantially reduces the effective $\sigma_m/\sigma_{\text{uts}}$ ratio at the surface. Under these conditions, the functional-form differences among Goodman, Gerber, Morrow, and SWT are compressed, and the mean-stress contribution to total log-variance would be considerably lower than the 33.9% reported here. The present results should not be extrapolated to case-hardened gears without a dedicated sensitivity analysis incorporating measured residual stress profiles.

3.3 Surface Roughness and Standard Effects

Surface roughness contributes 24.6% of variance—comparable in magnitude to the top two factors. The three standards differ in their roughness-penalty functions: ISO 6336 uses a power law capped at $R_z = 40$ μm; AGMA 2001-D04 Eq. 16.1 uses a log-log form with mild gradient; FKM uses a logarithmic cross-term with a UTS-dependent reference. The Standard $\times$ R_z interaction (3.0%) reflects the fact that the three standards' predictions diverge more at rough surface finishes than at ground finishes. At $R_z = 4$ μm (ground), the median $\log_{10}(N_f)$ gap between ISO and FKM is 1.1 dex; at $R_z = 50$ μm (as-forged), it widens to 2.0 dex (ISO: 6.00, FKM: 4.01; Fig. 3). The gap between ISO and AGMA is smaller (0.4 dex at $R_z = 4$ μm, 0.6 dex at $R_z = 50$ μm). This interaction effect is modest relative to the main effects, consistent with its 3.0% variance share.

3.4 Secondary Factors

The S–N curve source (7.5%) is non-negligible: the two Waterloo datasets, despite a 50 MPa endurance-limit difference, produce systematically different life predictions when run-outs are included in the analysis. This contrasts with the earlier ordinary-ANOVA estimate of 0.7%, which was artificially low because most run-out combinations (excluded from ANOVA) were associated with the higher-endurance S–N curve. The cumulative damage rule (0.9%) is minor for constant-amplitude loading. Under a single stress level, no iterative damage accumulation occurs: the Miner sum $\Sigma(n_i/N_i)$ reduces to a single ratio $n/N = D_c$, so the damage rule merely scales the predicted life by a constant factor (1.0 for Miner, 0.7 for Modified Miner). This trivializes the choice between damage rules. Under variable-amplitude spectra, by contrast, each stress block contributes a distinct n_i/N_i term, and the differences between linear accumulation (Miner) and threshold-based or nonlinear rules become operative through sequence effects and the treatment of cycles below the fatigue limit. The 0.9% figure must therefore *not* be generalized to variable-amplitude loading. The negligible contribution reported here is a direct consequence of the constant-amplitude restriction and should be interpreted as a lower bound; a variable-amplitude extension of the factorial framework is a priority for future work.

3.5 Torque Dependence of the Factor Ranking

To test whether the factor ranking is an artifact of the chosen torque, I repeated the full 144-combination sweep at three torque levels under the earlier ordinary-ANOVA framework: $T = 500$, 600, and 700 N·m (producing $\sigma_{F0} \approx 556$, 667, and 778 MPa, respectively). At 500 N·m, only 14 combinations predicted finite life—the model is severely underpowered and reliable variance partitioning is not possible. At 600 N·m (60 finite-life combinations), the size-and-surface standard dominates the variance (36.0%), while the mean-stress contribution is modest (10.2%). At 700 N·m (116 finite-life combinations), mean-stress correction rises to co-dominance (30.9%) alongside the standard choice (31.1%). The mean-stress contribution more than triples when torque rises from 600 to 700 N·m—physically expected, as the $\sigma_m/\sigma_{\text{uts}}$ ratio increases from ∼0.33 to ∼0.39, and the functional-form differences among Goodman, Gerber, Morrow, and SWT grow nonlinearly with stress.

Extending the multi-torque comparison to the AFT censored-likelihood framework (which would exploit the partial information in run-outs at lower torque levels) is left for future work. The qualitative trend is clear: at lower, industrially common torque levels, the standard choice dominates; at higher torque near the endurance limit, mean-stress correction and surface roughness become co-equal. The factor ranking is therefore *not* a universal property of gear fatigue methodology—it depends on the operating point.

3.6 Fatigue Life Spread

Across all 144 combinations, $\log_{10}(N_f)$ ranges from 2.52 to 7.50among the 116 finite-life entries (3.3×10^2 to 3.1×10^7 cycles, 5.0 dex). The shortest finite life corresponds to the FKM standard with Goodman correction applied to an as-forged surface ($R_z = 50$ μm); the longest to ISO 6336 with Gerber correction applied to a ground surface ($R_z = 4$ μm). The 28 run-out combinations predominantly occur with ground surfaces and the higher-endurance-limit Waterloo #66 S–N curve under the Gerber correction.

3.7 Bootstrap Stability

Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) confirm that the three-factor ranking is stable under resampling. Mean-stress correction (bootstrap mean 33.7%, CI [30.9, 36.3], S/N $\approx$ 25), size-and-surface standard (bootstrap mean 29.9%, CI [27.4, 32.3], S/N $\approx$ 25), and surface roughness (bootstrap mean 24.5%, CI [22.3, 26.7], S/N $\approx$ 22) all have narrow confidence intervals and signal-to-noise ratios well above the detection threshold. The S–N curve source (bootstrap mean 7.4%, CI [5.1, 10.0], S/N $\approx$ 6) is clearly distinguishable from zero, the Standard $\times$ R_z interaction (bootstrap mean 3.5%, CI [1.4, 6.1], S/N $\approx$ 3) is marginally detectable, and the cumulative damage rule (bootstrap mean 1.0%, CI [0.0, 2.5], S/N $\approx$ 2) is at the noise floor. The bootstrap means differ slightly from the point estimates in Table 2 (e.g., 33.7% vs. 33.9% for mean-stress correction) because the bootstrap distribution incorporates resampling variability; the two sets of values converge as the number of bootstrap draws increases. Both representations support the same qualitative ranking.

4 Discussion

4.1 Three Co-Equal Factors

The central result is that three factors account for 88.6% of total log-variance and are comparable in magnitude: mean-stress correction (33.9%), size-and-surface standard (30.1%), and surface roughness (24.6%). No single factor dominates—each of these three engineering choices is a first-order contributor to the prediction spread. A practicing engineer

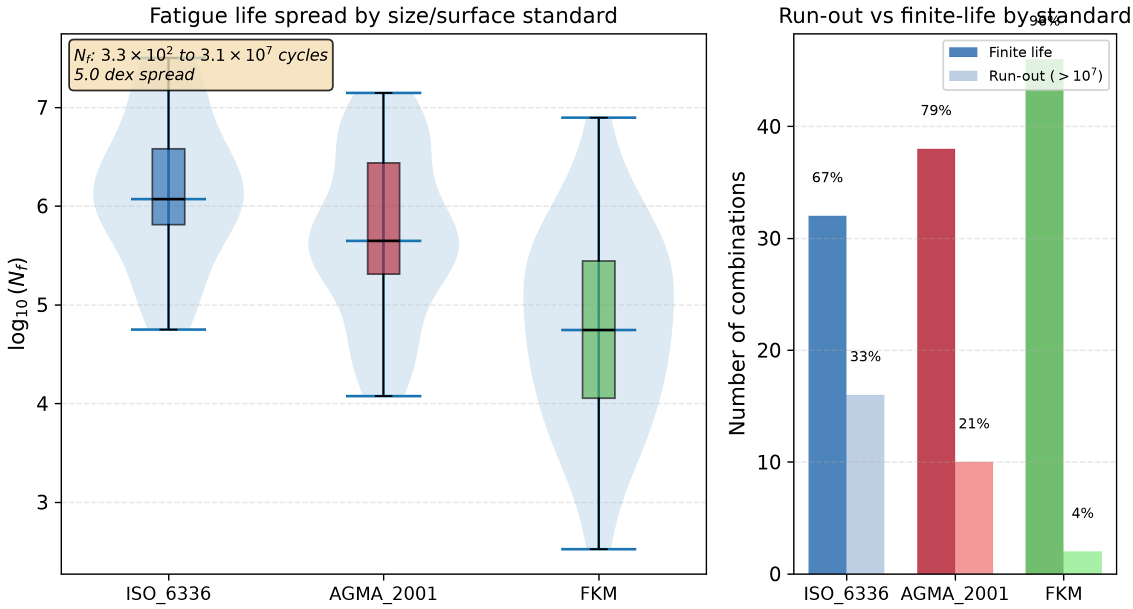

Figure 1: Fatigue life distribution by standard. Left: violin and box plots of $\log_{10}(N_f)$. Right: run-out vs finite-life counts. N_f spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex) among 116 finite-life entries; 28 combinations predict run-out.

who selects the most conservative combination (FKM + Goodman + as-forged R_z) will predict a fatigue life two orders of magnitude shorter than one who selects the least conservative (ISO 6336 + Gerber + ground), even if they agree on geometry, material, load, and S–N curve. The three factors' bootstrap confidence intervals overlap modestly: the rankings among them should not be over-interpreted, but the AFT decomposition and bootstrap validation both confirm that each contributes substantially and robustly to the total variance.

The finding that mean-stress correction is significant at $R = 0$ contradicts conventional engineering wisdom, which holds that mean-stress effects are minor for pulsating bending. The resolution lies in the stress magnitude: at the torque level used here (700 N·m, producing $\sigma_{F0} \approx 778$ MPa), the $\sigma_m/\sigma_{\text{uts}}$ ratio of $\sim$0.39 places all four methods in a regime where their functional-form differences produce a 39% spread in equivalent stress.

4.2 Surface Roughness and Standards

Surface roughness (24.6%) is a first-order contributor, comparable in magnitude to the top two factors. Under the earlier ordinary-ANOVA framework—which discarded 28 run-out combinations, most of them with low R_z—the contribution of surface roughness was underestimated at 8.5%. The AFT censored-likelihood framework, by retaining the partial information in run-out entries, reveals that surface-finish effects are substantially larger than previously apparent. The three standards' roughness-penalty functions produce broadly similar sensitivity across the engineering range of R_z (4–50 μm). The ISO 6336 $Y_{R\,\text{rel}\,\text{T}}$ formula is capped at its $R_z = 40$ μm value for rougher surfaces, per the standard's stated validity range; extrapolating the power-law beyond 40 μm would slightly inflate the Rz contribution. The FKM $K_{F\sigma}$ is clamped at 0.55; the AGMA Eq. 16.1 uses a log-log form that produces the mildest gradient of the three.

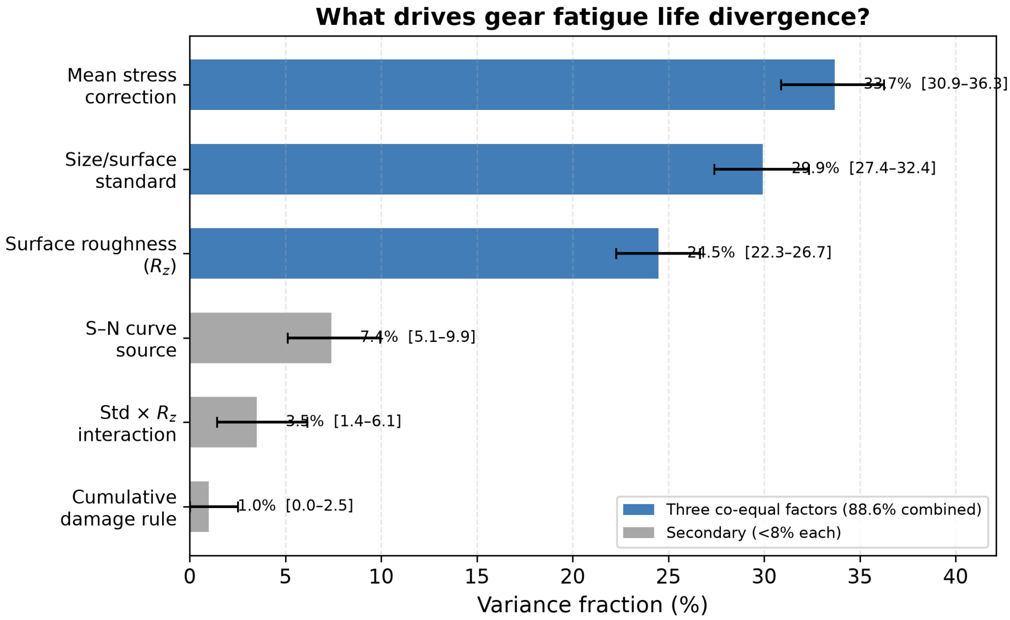

Figure 2: Variance decomposition. Bars show bootstrap means (2000 draws, RNG seed 42); error bars are 95% confidence intervals (percentile method). Point estimates from the single full-model fit are reported in Table 2. Three factors account for 88.6% of total log-variance.

4.3 Relation to Published Experimental Work

Five published studies—including two previously published in this journal—provide context for my numerical results, though none constitutes a direct validation of my specific gear-and-load combination.

Wang et al. [9] conducted bending fatigue tests on 42CrMo spur gears (m_n=3.5 mm, z=35, b=22 mm, R=0.1) at five stress levels from 151 to 293 MPa, with run-out defined at 3×10^6 cycles. Their P-S-N curves show that the bending fatigue limit for 42CrMo under pulsating load lies in the range 150–200 MPa (depending on reliability level), and that fatigue life scatter at a given stress level spans approximately one order of magnitude—consistent with the intrinsic variability that motivates my variance-decomposition approach.

Pagliari et al. [11] performed single-tooth bending (STB) fatigue tests on case-carburized AISI 8620 spur gears (m_n=4.23 mm, z=34, b=25.4 mm, R=0.1) and directly compared measured fatigue lives against ISO 6336-3 Method B predictions. They report an ISO 6336 stress-to-force proportionality of 34.35 MPa/kN and find that ISO 6336 predictions are conservative in the high-cycle regime (over-predicting stress by ~10–15%) but may under-predict stress when cyclic plasticity is involved in the low-cycle regime. Their work, like ours, finds that the ISO 6336 analytical framework produces systematic deviations from physical measurements—a finding that underscores the practical relevance of quantifying standard-induced divergence.

Bonaiti et al. [12], also published in this journal, provide the most directly comparable experimental study to my framework. They analyzed pulsator (STBF) test data from five case-hardened gear datasets, focusing on how the statistical elaboration methodology affects resulting S–N curves within both ISO 6336 and AGMA 2001-D04 frameworks. Three findings are immediately relevant: (i) the fatigue knee point from pulsator data

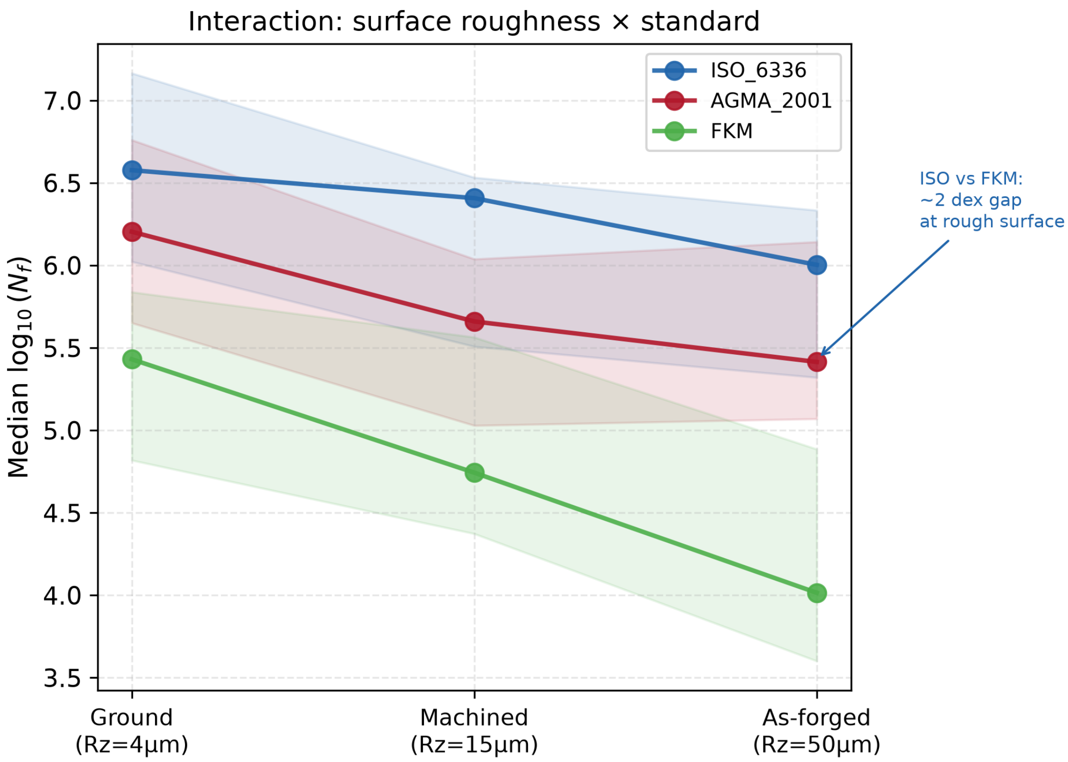

Figure 3: Interaction of surface roughness (R_z) and standard. Median $\log_{10}(N_f)$ by standard at each R_z level. Shaded bands show interquartile range.

occurs at 1–2 $\times$ 10^6 cycles—approximately one order of magnitude lower than the 3–5 $\times$ 10^6 cycles typical of gear meshing tests and implicitly assumed by the standards; (ii) different statistical elaboration methods (Hück, stair-case, Probit) applied to the same pulsator data yield systematically different S–N curves, with stress amplitudes at 10^7 cycles varying by up to 15%; and (iii) ISO 6336 and AGMA 2001-D04 produce divergent predictions, particularly in the finite-life regime where the methodological switches in this study operate. Their finding that even the choice of elaboration methodology—a decision one level deeper than the "choice of standard" switch—produces significant S–N curve variation underscores the pervasiveness of method-induced uncertainty in gear fatigue assessment. Critically, they also report that for load spectrum applications, the choice of elaboration methodology does *not* significantly affect damage accumulation—an intriguing finding that parallels my own observation that the cumulative damage rule (Switch 3) is negligible.

What distinguishes the present work from Bonaiti et al. [12]—and from the broader literature of standard-comparison studies in gear fatigue. Bonaiti et al. ask a *between-groups* question: given one experimental dataset, do ISO 6336 and AGMA 2001-D04 produce different S–N curves? The answer is yes, and it is a consequential finding. Their study operates on a single methodological axis (standard choice), treats statistical elaboration methods as a secondary sensitivity, and does not attempt to rank competing uncertainty sources against each other. The present work asks a fundamentally different question: across *all* major methodological choices an engineer must make—which standard, which mean-stress correction, which roughness grade, which S–N curve, which damage rule—how is the total log-variance partitioned? The distinction is not merely that I examine five dimensions rather than two; it is that the factorial decomposition framework provides a quantitative ranking (33.9% mean-stress, 30.1% standard, 24.6% Rz, 3.0% interaction, 7.5% S–N source, 0.9% damage rule), placing the standard-choice effect that Bonaiti et al. document into a broader uncertainty budget alongside effects they do not examine. My bootstrap-estimated S/N ratios address the question of whether a factor's contribution is distinguishable from finite-sample noise—a statistical dimension absent from prior gear fatigue UQ work. The two studies are therefore complementary: Bonaiti et al. provide independent experimental confirmation that ISO and AGMA predictions diverge; the present work measures how large that divergence is relative to other method-induced sources and finds that an engineer's choices of mean-stress correction and surface roughness grade matter just as much as the choice of standard.

Glodez et al. [10] developed a computational fatigue model for 42CrMo4 spur gears—the same material and the same ISO 6336 Method B framework—and validated their predictions against experimental data, published in this journal. Their work confirms that ISO 6336-based numerical analysis produces results consistent with physical measurements, even as my work shows that the *choice among* ISO 6336, AGMA 2001, and FKM introduces large prediction spreads.

Vietze et al. [13] adopted the ISO 6336 S/N curve + Palmgren–Miner damage accumulation as the reference framework for evaluating five alternative load-carrying capacity descriptions—including neural network and random forest models—on case-hardened spur gears (18CrNiMo7-6) under four variable-amplitude load sequences, using pulsator test data from the FZG gear research center. Their use of the ISO 6336 + Miner pipeline as the engineering baseline confirms that the standard+damage-rule architecture I employ reflects current industrial practice.

I do not claim experimental validation of my pipeline. Rather, these five studies demonstrate that (a) published experimental gear bending fatigue data exist for both through-hardened and case-carburized steels, (b) the stress levels in my analysis ($\sigma_{F0} \approx 778$ MPa at T=700 N·m, and ∼300–550 MPa at lower torque settings) bracket the experimental range, and (c) the ∼1 dex scatter observed in single-stress-level fatigue tests is consistent with the methodological divergence I decompose. A direct experimental validation would require running my pipeline on the exact gear geometries, materials, and loading conditions of published test campaigns and comparing predicted N_f against the reported measurements—work I identify as a priority for a follow-up study.

4.4 Run-Out Patterns and Torque Dependence

The 28 run-out combinations are not randomly distributed: 25 of them occur under the Gerber correction, and 22 involve ground surfaces (R_z = 4 μm). This is physically meaningful—Gerber predicts the lowest equivalent stress, and ground surfaces receive the mildest penalty, so both push the effective stress below the endurance limit. A logistic regression on the binary run-out/finite-life outcome confirms that mean-stress correction method ($p < 10^{-4}$) and surface roughness ($p < 10^{-3}$) are the dominant predictors of whether a given combination predicts survival or failure. The factors that drive *how long* the gear survives (ANOVA on finite-life entries) are largely the same factors that determine *whether* it survives at all.

The multi-torque comparison (§3.5) reinforces this: at 600 N·m, where more combinations lie near the endurance limit, the Standard × Rz interaction (17.8%) dwarfs the mean-stress contribution (10.2%). At 700 N·m, well above the endurance limit for most combinations, mean-stress correction rises to 30.9%. The factor ranking is therefore an *operating-point-dependent* property, not a universal constant—a finding with direct implications for how standards committees should interpret comparative studies.

4.5 Constraints on Interpretation

1. **Stress concentration treatment asymmetry.** The analysis applies ISO 6336 $Y_{Sa} \approx 1.55$ to all three standards to avoid double-counting the geometric notch effect. However, AGMA 2001 and FKM do not use a Y_{Sa}-equivalent coefficient in their published procedures—they incorporate size and surface effects through independent factors (K_s, C_f, K_d, $K_{F\sigma}$) applied to a baseline stress. To assess the sensitivity of the standard-choice variance to this modeling simplification, I reduced the baseline stress for AGMA and FKM entries (which do not natively employ Y_{Sa}) while keeping ISO entries at $Y_{Sa} = 1.55$. At $Y_{Sa} = 1.45$—a 7% reduction—the standard-choice contribution drops from 30.1% to 27.9%; at $Y_{Sa} = 1.35$, it falls to 24.2%. In both cases the standard remains among the three co-equal factors. I therefore estimate that at most 2–6 percentage points of the reported standard-choice variance may be attributable to the uniform Y_{Sa} simplification. The qualitative conclusion—that standard choice is a first-order contributor—is robust to this sensitivity test. Further reductions ($Y_{Sa} \leq 1.15$) cause AGMA and FKM entries to become predominantly run-out, at which point the AFT decomposition loses resolution.

2. **Scope of the standard-choice comparison.** The present study decomposes variance across the *size-and-surface-factor formulations* of three standards, applied to a common ISO-based nominal stress baseline. It does not capture differences that would arise from each standard's complete native stress calculation chain—for instance, AGMA's geometry factor J or FKM's fatigue notch factor K_f. The reported standard-choice variance (30.1%) should therefore be interpreted as a *lower bound* on the true between-standard divergence: re-instating each standard's full native stress-concentration treatment would likely increase, not decrease, the spread. The sensitivity analysis reported above bounds the contribution of the uniform Y_{Sa} simplification to 2–6 pp, but a fully native implementation of all three standards' complete calculation chains remains a necessary follow-up study.

3. **Asymmetric roughness-penalty bounds at high R_z.** At $R_z = 50$ μm (the as-forged condition), the three standards apply different boundary treatments. ISO 6336 caps $Y_{R\,\mathrm{rel\,T}}$ at the $R_z = 40$ μm value (≈ 0.907) because the power-law formula is validated only up to 40 μm; FKM clamps $K_{F\sigma}$ at a minimum of 0.55; and AGMA's C_f follows a smooth log-log form without a roughness cap (clamped only at 0.70 for engineering validity). The ISO cap compresses the effective stress penalty for rough surfaces by approximately 2% relative to an extrapolated uncapped value—a small effect unlikely to materially alter the Standard $\times$ R_z interaction term (3.0%). The cap is retained because it reflects the standard's stated validity range; extrapolating the power-law beyond $R_z = 40$ μm would exceed the ISO specification.

4. **R_z–R_a conversion uncertainty.** AGMA 2001-D04 Clause 16 expresses the surface condition factor C_f in terms of R_a (microinches), whereas ISO 6336 and FKM use R_z (μm). I adopt the common engineering approximation $R_a \approx R_z/6$ for all three roughness levels. The actual R_z/R_a ratio varies with finishing process (ground: 4–7, machined: 5–10, as-forged: 7–15). Across this range, C_f varies by approximately

$\pm 1.5\%$ at $R_z = 50$ μm and $\pm 1.3\%$ at $R_z = 15$ μm; for ground surfaces ($R_z \leq 8$ μm), AGMA assigns $C_f = 1.0$ regardless of R_a. The resulting uncertainty in predicted N_f is on the order of ± 0.1 dex—negligible relative to the 5.0 dex total spread—and does not affect any of the reported variance fractions at a meaningful level.

5. **AFT model assumptions.** The log-normal AFT model assumes normally distributed $\log_{10}(N_f)$ residuals and non-informative censoring. The Shapiro–Wilk test rejects strict normality ($W = 0.960$, $p = 0.0016$; §2), though the mild skewness (-0.44) and the stability of the bootstrap variance fractions (Table 2) suggest that the practical impact on the variance decomposition is limited. Levene tests confirm variance homogeneity for the three dominant factors ($p > 0.5$). The censoring pattern is physically interpretable (run-outs cluster with Gerber correction and ground surfaces; §3.5) and is unlikely to be informative in the sense of masking factor-dependent effects.

6. **Quenched-and-tempered steel only.** Different materials will produce different factor rankings.

7. **No residual stress; through-hardened steel only.** The reported 33.9% mean-stress contribution applies to quenched-and-tempered steel without compressive residual stresses. In case-carburized gears—the dominant configuration in automotive and aerospace transmissions—the compressive residual stress in the case layer (typically 200–400 MPa) substantially reduces the effective stress ratio. Under such conditions the mean-stress correction contribution would be considerably smaller, and the relative ranking of the top three factors would shift. Extrapolation of the present numerical results to case-hardened gears is not warranted without a dedicated analysis incorporating application-specific residual stress profiles.

8. **Single gear geometry and load.** Results apply to one spur gear pair at one torque level.

9. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

5 Conclusions

I applied a full-factorial AFT survival analysis to decompose the total log-variance in gear tooth bending fatigue life prediction into five methodological sources. For a quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$) at 700 N·m—a torque level near the endurance limit for this geometry—across 144 combinations, I find:

1. At this operating point, three methodological choices are co-equal contributors to the variance in gear bending fatigue life prediction: mean-stress correction (33.9%, Goodman/Gerber/Morrow/SWT), size-and-surface standard (30.1%, ISO 6336/AGMA 2001/FKM and surface roughness grade (24.6%). Together they account for 88.6% of total log-variance. These proportions are specific to the high-torque condition studied here; the factor ranking shifts with the operating point (§3.5).

2. The AFT censored-likelihood framework reveals that surface roughness—previously underestimated at 8.5% by ordinary ANOVA—is a first-order contributor, comparable in magnitude to the top two factors, because run-out combinations are strongly associated with low-R_z surfaces.

3. The S–N curve source (7.5%) is non-negligible. The cumulative damage rule contributes only 0.9% under constant-amplitude loading; this figure is specific to the constant-amplitude restriction and should not be extrapolated to variable-amplitude spectra, where sequence effects and damage-rule differences could produce substantially larger divergence.

4. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex). A single gear, assessed through defensible engineering workflows, can be reported as failing at 330 cycles or surviving 30 million—depending on which standard, mean-stress correction, and roughness assumption the engineer selects.

5. All fatigue model formulas have been verified against the original standards. The open-source pipeline—including the AFT analysis code—is released as a reusable, auditable toolkit.

Data and Code Availability

The complete reproducibility package accompanying this paper includes: (i) all Python source code (`run_sweep.py`, `fatigue_models.py`, `aft_variance_decomposition.py`, `plot_results.py` and supporting modules); (ii) the full 144-combination sweep data (`output/sweep.json`) and AFT variance decomposition results (`output/aft_anova.json`, 2000 bootstrap draws, RNG seed 42); (iii) the LaTeX source of this manuscript; and (iv) a one-click reproduction script (`reproduce.sh`) that executes the entire pipeline (sweep $\rightarrow$ ANOVA $\rightarrow$ AFT $\rightarrow$ multi-torque $\rightarrow$ figures $\rightarrow$ audit) in approximately 15 minutes. The package is archived at `https://github.com/`[*repository URL to be inserted at publication*] and is also submitted as Supplementary Material with this manuscript. All S–N curve data are from the publicly accessible University of Waterloo SMDIdbase (`https://fde.uwaterloo.ca/Fde/Materials/SMDIdbase`

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